# An Artificial Intelligence-Based Classification and Ranking System for Border Gateway Protocol Incidents in Enterprise Networks

by Biruk D. Mekonnen

B.S. in Electrical Engineering, July 2007, Mekelle University

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Praxis directed by

Timothy Blackburn, PhD

Professorial Lecturer of Engineering Management and Systems Engineering

Amir Etemadi, PhD

Assistant Professor of Engineering and Applied Science

The School of Engineering and Applied Science of The George Washington University certifies that Biruk D. Mekonnen has submitted this Praxis for review in partial fulfillment of the requirements for the Doctor of Engineering degree. Final committee signature details should be completed using the official approved form.

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Biruk D. Mekonnen

Praxis Research Committee:

Dr. Amir Etemadi, Assistant Professor of Engineering and Applied Science, Praxis Director

Dr. Timothy Blackburn, Professorial Lecturer of Engineering Management and Systems Engineering, Praxis Co-director

[Third committee member name and title to be confirmed], Committee Member

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# Dedication

[Dedication text to be finalized by the author.]

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# Abstract of Praxis

An Artificial Intelligence-Based Classification and Ranking System for Border Gateway Protocol Incidents in Enterprise Networks

Border Gateway Protocol (BGP) incidents can disrupt reachability, traffic engineering, and operational confidence across interdomain networks. Prefix hijacks, route leaks, and outages may create overlapping symptoms in public routing telemetry, making it difficult for operators to quickly prioritize the most likely incident cause. This Praxis develops and evaluates an artificial intelligence-based system that classifies and ranks three BGP incident categories using historical RouteViews UPDATE data, incident-level labels, and routing-security context.

The final pipeline uses a frozen corpus of 1,419 independent incidents, balanced across 473 hijack, 473 leak, and 473 outage incidents. All splits were performed at the incident level: 993 incidents for TRAIN, 213 for Validation, and 213 for the untouched Final Holdout. Five-minute windows were represented with 74 frozen V4 features: 44 behavioral BGP features, 14 CAIDA AS-relationship features, and 16 RPKI/ROV features.

Model development began with baseline tree-based classifiers and progressed through imbalance treatment, feature-variant ablation, feature-family ablation, Random Forest tuning, XGBoost refinement, and a final incident-level stacking architecture. The terminal model, A0_XGB_TO_XGB K1, combines window-level XGBoost probability evidence with incident-level temporal summary features and a second-level XGBoost classifier.

On the untouched Final Holdout of 213 incidents, the final system achieved incident-level Macro F1 of 0.920177 with a 95% confidence interval of [0.880117, 0.954402]. Top-1 accuracy was 0.920188, Top-2 accuracy was 0.990610, and mean reciprocal rank was 0.958529. The true class was ranked in the Top-2 for 211 of 213 incidents. Per-class F1 scores were 0.877698 for Hijack, 0.904110 for Leak, and 0.978723 for Outage.

The study demonstrates that combining behavioral BGP features with CAIDA relationship context and RPKI/ROV origin-validation signals can support accurate incident-level classification and ranked incident-cause output. The evidence also shows important limitations: public RouteViews collectors do not fully represent enterprise-local routing state, incident labels depend on external sources, and the Normal-vs-Incident Stage 1 formulation remained experimental rather than part of the final architecture.

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# List of Symbols

Table S.1 Key Symbols and Abbreviations

| Symbol | Meaning |
|---|---|
| AS | Autonomous System |
| AS_PATH | Ordered AS sequence in a BGP route announcement |
| BGP | Border Gateway Protocol |
| F1 | Harmonic mean of precision and recall |
| MRR | Mean reciprocal rank |
| NLRI | Network Layer Reachability Information |
| ROA | Route Origin Authorization |
| ROV | Route Origin Validation |
| RPKI | Resource Public Key Infrastructure |

# List of Acronyms

AS Autonomous System

AS_PATH Ordered AS sequence in a BGP route announcement

BGP Border Gateway Protocol

F1 Harmonic mean of precision and recall

MRR Mean reciprocal rank

NLRI Network Layer Reachability Information

ROA Route Origin Authorization

ROV Route Origin Validation

RPKI Resource Public Key Infrastructure

# Chapter 1-Introduction

## Background and Research Motivation

The Internet depends on BGP to exchange reachability information among autonomous systems. Because BGP was designed around distributed trust and local routing policy, abnormal routing events can propagate quickly and may be difficult to interpret from control-plane evidence alone. Three incident categories are central to this Praxis: prefix hijacks, route leaks, and outages. Each can affect reachability or traffic paths, yet the observable symptoms in BGP UPDATE data may overlap.

Enterprise network teams and network operations centers typically rely on monitoring dashboards, route collectors, alerts, and engineer judgment to interpret routing anomalies. These tools may show that routing behavior changed, but they often provide limited support for ranking the most likely incident type from routing evidence. This creates a gap between anomaly detection and operator-facing incident interpretation.

This Praxis addresses that gap by treating incident classification and ranking as an empirical machine-learning problem. The study does not claim to infer private enterprise-local root causes from public telemetry. Instead, it asks whether a rigorously labeled, leakage-controlled corpus of public BGP observations can support reliable classification and ranked incident-cause output for hijack, leak, and outage incidents.

## Problem Statement

BGP incidents can be operationally disruptive, but their symptoms are often ambiguous in public routing telemetry. Existing approaches frequently focus on detecting anomalies or on one specialized incident type. Operators need a reproducible method that combines BGP behavioral evidence, AS-relationship context, and route-origin validation signals to classify and rank likely incident categories without leaking future evaluation information.

## Thesis Statement

An incident-level machine-learning system that combines BGP behavioral features, CAIDA AS-relationship context, and RPKI/ROV route-origin validation signals can

accurately classify and rank hijack, leak, and outage incidents when trained and evaluated under leakage-safe incident-level partitions.

## Research Objectives

- Construct a balanced, quality-controlled, incident-level BGP corpus for hijack, leak, and outage events.
- Extract reproducible five-minute behavioral features from historical RouteViews UPDATE data.
- Enrich the behavioral representation with CAIDA AS-relationship and historical RPKI/ROV features.
- Evaluate model families and feature variants using incident-grouped validation to prevent leakage across windows from the same incident.
- Develop and freeze a final incident-level classification and ranking model before a one-time Final Holdout evaluation.
- Quantify classification, ranking, uncertainty, limitations, and future operational-readiness requirements.

## Research Questions and Hypotheses

Table 1.1 Research Questions and Evidence

| ID | Question | Evidence Basis |
|---|---|---|
| RQ1 | Can historical RouteViews data support a reproducible BGP incident feature corpus? | Supported by a frozen TRAIN behavioral corpus of 68,031 five-minute rows and a V4 feature schema. |
| RQ2 | Do CAIDA and RPKI enrichments improve incident classification? | Supported with caveat by V1-V4 feature variants and feature-family ablation. |
| RQ3 | Which architecture best classifies and ranks incident causes? | Supported by TRAIN grouped-CV comparisons, K1 stacking, K2 stability checks, and K2.5 architecture comparison. |
| RQ4 | Does the locked final system generalize to unseen incidents? | Supported by one-time Final Holdout evaluation on 213 |

| | | |
|---|---|---|
| | | incidents. |

## Research Scope and Limitations

The empirical scope is restricted to labeled hijack, leak, and outage incidents observable through public RouteViews collectors and enriched with publicly available CAIDA and RPKI data. The system classifies and ranks incident categories; it does not prove causal mechanisms or replace operator forensics. Validation and Final Holdout partitions were separated at the incident level, and the Final Holdout remained sealed until the terminal evaluation.

## Praxis Organization

- Chapter 1 introduces the motivation, problem, objectives, research questions, scope, and limitations.
- Chapter 2 reviews BGP measurement, incident types, RPKI, AS relationships, and machine-learning approaches to routing anomalies.
- Chapter 3 describes the incident corpus, RouteViews processing, feature engineering, enrichment, modeling, and evaluation protocol.
- Chapter 4 presents the experimental results, feature ablations, model comparisons, validation evidence, K1 architecture, and Final Holdout findings.
- Chapter 5 maps findings to the research questions, states contributions and limitations, and identifies future research.

# Chapter 2-Literature Review

## Introduction

The literature supporting this Praxis spans BGP operation, routing anomalies, RPKI, AS relationships, public measurement archives, machine learning for network anomalies, and ranking-oriented evaluation. The central theme is that BGP incidents are not only detection problems; once an abnormal routing event is observed, operators benefit from evidence that helps prioritize plausible incident categories.

Prior work has produced important specialized systems for hijack detection, leak detection, anomaly detection, and routing-security measurement. This Praxis builds on that foundation by integrating behavioral, AS-relationship, and RPKI signals into a single incident-level classification and ranking framework.

## Foundational Studies on BGP Operation and Measurement

BGP-4 defines the interdomain routing protocol used to exchange reachability among autonomous systems (Rekhter et al., 2006). Route selection depends on path attributes, local policy, and incremental UPDATE messages. This design supports Internet-scale policy autonomy, but it also means global behavior emerges from many local decisions rather than from centralized validation.

Public collector projects such as RouteViews and BGPStream provide a practical foundation for historical BGP analysis (University of Oregon, 2000; Orsini et al., 2016). These archives make it possible to reconstruct observed announcements, withdrawals, AS paths, peer visibility, and temporal changes around incidents, while also imposing an important limitation: collector visibility is partial and does not equal enterprise-local telemetry.

## BGP Routing Incidents: Hijacks, Leaks, and Outages

Prefix hijacks occur when an unauthorized or unexpected AS originates reachability for a prefix. Prior systems such as PHAS, Argus, ARTEMIS, and forged-origin hijack detectors show that hijack detection can use origin changes, prefix relationships, and

validation evidence (Lad et al., 2006; Shi et al., 2012; Sermpezis et al., 2018; Holterbach et al., 2024).

Route leaks occur when routes are propagated beyond intended policy boundaries. The literature emphasizes AS-path relationships, role violations, valley-free routing, and leak localization (Azimov et al., 2022; Bagnulo et al., 2022; Li et al., 2023; Xie et al., 2025). These studies motivate CAIDA relationship features and valley-free indicators in this Praxis.

Outages differ from hijacks and leaks because the operational event is often a loss of reachability or service availability rather than a suspicious path assertion. In BGP data, outages may appear through withdrawals, reduced peer visibility, or instability for affected ASNs and prefixes. This distinction matters for both feature design and label interpretation.

## RPKI, ROA, and Route Origin Validation

RPKI provides a cryptographic framework for expressing which AS is authorized to originate a prefix (Lepinski & Kent, 2012). Route Origin Authorizations and Route Origin Validation can classify observed prefix-origin pairs as valid, invalid, or unknown (Mohapatra et al., 2013; Bush, 2014).

RPKI deployment and operational behavior are uneven across the Internet, so invalidity is not a complete incident explanation by itself (Chung et al., 2019; Rodday et al., 2024). For this Praxis, RPKI is treated as a predictive feature family rather than as a sole label generator. This is especially important for hijack classification, where origin authorization evidence can be highly informative but not universally available.

## AS Relationships, Routing Policy, and Valley-Free Routing

AS-relationship inference has long been used to characterize routing policy and path economics (Gao, 2001; Gao & Rexford, 2001; Dimitropoulos et al., 2007; Luckie et al., 2013). Customer, provider, and peer relationships provide a policy context for interpreting observed AS paths.

Valley-free assumptions are not perfect, but violations and unusual relationship sequences are useful evidence for route-leak analysis. The CAIDA feature family in this Praxis operationalizes this literature by measuring relationship counts, peer/provider/customer transitions, valley-free violations, and related path-context signals.

## Machine Learning for BGP Anomaly Detection and Incident Classification

Machine-learning studies have used support vector machines, decision trees, Random Forests, gradient boosting, recurrent neural networks, graph representations, and hybrid robust-statistical methods for BGP anomalies (Al-Rousan & Trajković, 2012; Sanchez et al., 2019; Romo-Chavero et al., 2025). These works show that engineered routing features remain competitive for structured BGP data.

Tree-based methods are particularly appropriate for tabular feature representations. Random Forest provides a strong nonlinear baseline (Breiman, 2001), while XGBoost often performs strongly on tabular classification tasks through sequential boosting and regularization (Chen & Guestrin, 2016). The final system in this Praxis uses XGBoost in both window-level and incident-level stages.

The literature also motivates careful evaluation. Random row-level splitting can leak information when many windows originate from the same incident. This Praxis therefore uses incident-level partitions and grouped cross-validation so that windows from one incident cannot appear in both training and evaluation folds.

## Explainability, Ranking, and Operational Interpretation

Operational incident support requires more than a single class label. Ranking metrics such as Top-2 accuracy and mean reciprocal rank evaluate whether the true incident category appears near the top of the model's ordered output. These measures fit operator workflows in which the first investigation step may not always be correct, but a short ranked list can still reduce search effort.

Explainability methods such as SHAP can help interpret which features influence predictions (Lundberg & Lee, 2017; Lundberg et al., 2020). However, feature attribution does not prove causality. This Praxis treats explanations as diagnostic aids and evidence checks, not as proof that a model discovered the real-world root cause.

## Summary of Research Gaps and Study Positioning

The literature leaves several gaps that this Praxis addresses. First, many systems focus on one incident type, while operational triage often requires comparison among multiple plausible incident categories. Second, routing-security signals and AS-relationship signals are often studied separately rather than integrated into one frozen feature corpus. Third, evaluation frequently risks leakage when windows rather than incidents define the independent unit.

This study positions its contribution as an engineering and empirical validation contribution: a reproducible incident-level corpus, a frozen Behavioral+CAIDA+RPKI feature representation, leakage-safe model development, ranked incident-cause output, and a terminal Final Holdout evaluation.

## Detailed Literature Review from Current Manuscript

The following section preserves and updates the broader literature-review material already developed in the current Praxis manuscript. Terminology has been aligned with the corrected study scope: classification and ranked incident-cause output rather than unsupported causal root-cause proof.

## Introduction

The Border Gateway Protocol (BGP) is the principal interdomain routing protocol through which autonomous systems (ASes) exchange reachability information and implement routing policy across the global Internet. Its path-vector design allows independently administered networks to select and propagate routes according to local economic, engineering, and security objectives rather than according to a single global optimization criterion (Rekhter et al., 2006; Gao, 2001). This flexibility made BGP scalable enough to support the growth of the Internet, but it also preserved a trust model

in which route announcements were historically accepted without cryptographic proof that the announcing AS was authorized to originate a prefix or that the advertised AS path reflected an authorized propagation sequence. Consequently, routing instability, configuration errors, route leaks, and prefix hijacks remain important sources of operational risk.

The literature on BGP has developed along several complementary lines. Foundational studies characterize protocol operation, convergence, routing policy, and the measurement of global control-plane behavior. A second body of work documents incidents and outages and demonstrates how local configuration errors or unauthorized advertisements can propagate far beyond the network in which they originate. A third line of research focuses on routing security mechanisms, particularly the Resource Public Key Infrastructure (RPKI), Route Origin Authorizations (ROAs), Route Origin Validation (ROV), and BGPsec. A fourth examines AS-level business relationships and the valley-free routing model, which provide semantic information about whether an observed propagation path is consistent with expected customer-provider and peer-to-peer policies (Gao, 2001; Luckie et al., 2013; Giotsas et al., 2014).

These security and policy perspectives are especially relevant to anomaly interpretation. An origin change, for example, is more informative when evaluated against RPKI validity than when treated only as a count of changed origins. Similarly, an unusual AS path becomes more meaningful when its adjacent AS relationships and valley-free behavior are considered. RPKI and AS-relationship data therefore provide contextual features that can distinguish superficially similar control-plane patterns. This distinction is important because a route leak, a hijack, a session reset, and ordinary convergence can all produce bursts of BGP UPDATE messages, path changes, or withdrawals while arising from materially different causes (Bagnulo et al., 2022; Xie et al., 2025; Rodday et al., 2024).

Machine-learning research has increasingly attempted to exploit these observable signals. Studies have applied support vector machines, hidden Markov models, decision trees, Random Forests, gradient-boosted models, clustering, recurrent neural networks, graph representations, and graph attention networks to BGP anomaly detection and

classification (Al-Rousan & Trajković, 2012; Sanchez et al., 2019; Dong et al., 2021; Peng et al., 2021; Romo-Chavero et al., 2025). More recent systems move toward semantics-aware analysis, topology-aware representations, and incident localization rather than simple threshold-based detection (Chen et al., 2024; Xie et al., 2025; Latif-Martínez et al., 2025). Explainable artificial intelligence (XAI), including SHAP-based feature attribution, further creates an opportunity to connect model predictions to routing behaviors that network engineers can inspect and validate (Kyryk et al., 2025).

Despite this progress, the literature remains fragmented. Security mechanisms generally validate a specific property of a route; anomaly-detection systems frequently optimize the identification of abnormal behavior; and route-leak or hijack systems are often specialized for one incident class. Comparatively few studies combine behavioral BGP features with both origin-validation evidence and AS-policy semantics and then use those signals to rank competing incident explanations. The distinction between detection and diagnosis is consequential in enterprise operations, where an alert that routing behavior is abnormal does not by itself tell an operator whether the most likely cause is a hijack, route leak, session reset, or another routing disturbance.

This chapter reviews the literature that supports the design of an artificial intelligence–based classification and ranking system for BGP incidents in enterprise networks. Section 2.2 reviews foundational BGP operation and measurement. Section 2.3 examines routing incidents and outages. Section 2.4 evaluates BGP security mechanisms, with particular emphasis on RPKI, ROA, ROV, and BGPsec. Section 2.5 reviews AS relationships, routing policies, valley-free routing, and CAIDA-style relationship inference. Section 2.6 synthesizes BGP anomaly-detection techniques. Section 2.7 evaluates statistical, ensemble, deep-learning, graph-based, and explainable-AI approaches for network fault detection and diagnosis. Sections 2.8 and 2.9 identify the research gaps and synthesize the literature into the model justification used by this praxis.

## Foundational Studies on BGP Operation and Measurement

BGP emerged as the Internet evolved from a comparatively simple hierarchy into a federation of independently operated networks. RFC 4271 defines BGP-4 as a path-

vector protocol in which routes are associated with path attributes and exchanged over long-lived TCP sessions (Rekhter et al., 2006). Unlike an interior gateway protocol that attempts to optimize a common metric within one administrative domain, BGP allows each AS to express local policy. The resulting route is therefore not necessarily the shortest topological path; it is the path that survives import policy, local preference, path selection, and export policy across a chain of independently controlled networks.

BGP UPDATE messages provide the principal observational unit for many routing studies. An UPDATE can announce Network Layer Reachability Information (NLRI), withdraw previously advertised reachability, and carry path attributes such as AS_PATH, ORIGIN, NEXT_HOP, MULTI_EXIT_DISC (MED), and communities. Of these, AS_PATH is particularly important to anomaly research because it exposes the sequence of AS numbers associated with an advertisement. Changes in path length, path composition, origin AS, or the frequency with which paths change can therefore be transformed into measurable behavioral features. Communities add another policy signal because operators use them to communicate routing intent, traffic-engineering preferences, and propagation controls (Giotsas et al., 2018; Fezeu et al., 2022).

BGP's decision process also explains why control-plane symptoms can be ambiguous. A route may change because of a physical failure, a BGP session transition, a policy modification, a newly preferred path, a more-specific advertisement, or an unauthorized origin. Withdrawals can trigger path exploration, during which routers successively advertise alternate paths before reaching a stable state. Classic work on Internet routing instability demonstrated that convergence can generate substantial update activity even when no malicious event has occurred (Labovitz et al., 2000). For anomaly-detection research, this means that update volume alone is an incomplete indicator: benign convergence and security incidents can occupy overlapping regions of the observable feature space.

Empirical BGP research depends heavily on route collectors. RouteViews and the RIPE NCC Routing Information Service (RIS) maintain distributed collectors that receive routes from participating networks and archive routing table snapshots and UPDATE streams in MRT-compatible formats. These archives enable retrospective reconstruction

of incidents and longitudinal measurement of route visibility. BGPStream provides an abstraction layer for accessing and processing such data, while tools such as BML have been developed to simplify the construction of BGP datasets for machine-learning experiments (Hoarau et al., 2021). The availability of public control-plane archives has therefore transformed BGP research from small laboratory experiments into studies based on Internet-scale observations.

Route collector data nevertheless represent a sampled view of the routing system rather than a complete ground truth. Visibility depends on collector placement, participating peers, routing policy, and the routes exported to each monitor. A route visible at one collector may be absent or may appear at a different time at another. This limitation matters when building event windows and comparing incidents. The present study uses RouteViews consistently for BGP UPDATE collection; the methodological rationale and observed archive-timing considerations that led to that choice are described in Chapter 3. In the literature review, the important point is that both RouteViews and RIS are measurement infrastructures whose observations are shaped by vantage-point coverage and collector behavior.

Dataset construction is itself a substantive research problem. Fonseca et al. (2019) and Hoarau et al. (2021) illustrate the need to transform raw routing records into structured observations that can support statistical or machine-learning analysis. Typical derived variables include announcement and withdrawal counts, ratios, unique prefixes, unique ASes, AS-path length statistics, origin changes, and temporal rates. Later studies extend this representation with graph structure, AS-level dynamics, or semantics-aware features (Peng et al., 2021; Liu et al., 2023; Latif-Martínez et al., 2025). These approaches establish the methodological precedent for treating public BGP telemetry as a source from which behavioral features can be engineered.

A further foundational issue is the AS-level topology. The Internet's AS graph is neither random nor static. Siganos et al. (2003) showed structural regularities in AS-level topology, while Gao (2001) demonstrated that the interpretation of AS links requires business relationships, not merely graph adjacency. Consequently, two identical-length AS paths may have very different policy meaning. This insight motivates the later

integration of AS-relationship information into anomaly features: path structure describes where a route traveled, whereas relationship semantics help describe whether that propagation is economically and operationally plausible.

Taken together, foundational BGP research provides three building blocks for this praxis. First, the protocol exposes observable behavioral signals through UPDATE messages and path attributes. Second, public collectors make those signals available at Internet scale. Third, AS-level topology and policy determine how those observations should be interpreted. These foundations support a feature-engineering strategy that combines raw behavioral statistics with external security and relationship context rather than treating BGP telemetry as an undifferentiated time series.

## BGP Routing Incidents and Outages

The operational consequences of BGP's distributed trust and policy model are visible in a long history of routing incidents. These events range from benign session resets and convergence episodes to accidental route leaks, prefix hijacks, and large-scale outages. A central lesson of the literature is that similar control-plane symptoms can emerge from different underlying causes. This many-to-one relationship between cause and observation is precisely what makes incident-cause diagnosis more difficult than anomaly detection.

Routing instability is often associated with bursts of announcements and withdrawals. When a preferred route disappears, BGP speakers may explore alternatives, generating transient paths and repeated updates before convergence. Such behavior can resemble an attack because both produce abrupt deviations from a baseline. Early studies of BGP instability therefore remain relevant to modern machine learning: they warn that a model trained only to recognize high update volume may confuse normal protocol dynamics with an incident (Labovitz et al., 2000).

Misconfiguration is another persistent source of routing disruption. Incorrect prefix advertisements, missing filters, inappropriate export policies, and unintended preference changes can cause routes to propagate outside their intended scope. A route leak is particularly instructive because the origin may be legitimate while the propagation is not.

Bagnulo et al. (2022) emphasize this distinction in ASIRIA: leaked routes frequently contain legitimate origins and legitimate individual AS hops but violate the intended scope implied by inter-AS relationships. This makes origin-only validation insufficient for many leak scenarios and motivates policy-aware features.

Prefix hijacking differs from a route leak in that an AS announces address space for which it is not an authorized origin, or an attacker constructs a forged-origin path designed to preserve the legitimate origin while manipulating propagation. ARTEMIS demonstrated that origin changes and control-plane visibility can support rapid detection and mitigation of many hijacks (Sermpezis et al., 2018). More recent work such as DFOH addresses forged-origin attacks that can evade simple origin-change checks (Holterbach et al., 2024). The evolution from straightforward origin hijacks to more sophisticated forged-origin scenarios shows why no single feature can reliably characterize the full space of BGP security incidents.

Route leaks and hijacks can also interact with more-specific routing. Because longest-prefix matching occurs in the forwarding plane, an unauthorized more-specific route may attract traffic even when a covering aggregate remains correctly originated. RPKI can mitigate many such cases when ROAs accurately authorize the expected origin and maximum prefix length, but invalid states can also arise from operational mistakes rather than attacks (Xu et al., 2019). Therefore, RPKI invalidity is strong evidence but should be interpreted in combination with other features rather than treated as a perfect incident label.

Operational outages add another class of ambiguity. A BGP session reset or loss of connectivity to an upstream may generate withdrawals, path substitutions, and changes in reachability without any malicious routing announcement. At Internet scale, the resulting control-plane signature can overlap with policy changes or route-leak recovery. For enterprise networks, session instability is especially relevant because multihomed edge designs depend on BGP sessions to maintain redundant external reachability. A diagnostic model must therefore recognize withdrawal-oriented and path-instability patterns without automatically equating them with hijacking.

The incident literature also demonstrates the importance of event-level context. A single UPDATE is rarely sufficient to determine cause. Instead, investigators reconstruct a time window, compare pre-event and in-event behavior, inspect path changes, identify affected prefixes and ASes, and consult external evidence. Systems such as BEAR seek to automate parts of BGP event analysis and reporting, while semantics-aware systems such as BEAM move toward explaining the structure of routing anomalies rather than merely raising alerts (Chen et al., 2024; Li et al., 2025). These directions are closely aligned with the operational goal of reducing the manual reasoning burden on network engineers.

For this study, the literature on incidents leads to a feature-level hypothesis: hijacks, leaks, and session resets should differ not necessarily in one isolated statistic but in combinations of behavioral, origin-validation, and AS-policy signals. Hijacks are expected to be particularly sensitive to origin changes and RPKI validity; leaks to relationship sequences and valley-free violations; and session resets to withdrawal and path-instability patterns. The literature does not imply that these signals are deterministic. Rather, it supports their use as evidence that can be learned and combined by a probabilistic classifier.

## BGP Security Mechanisms and Routing Validation

BGP's original specification does not cryptographically validate either the right of an AS to originate a prefix or the authenticity of the AS path carried in an UPDATE. Routing security research has therefore developed mechanisms that add verifiable information to the interdomain routing system. The most operationally significant are the RPKI ecosystem and ROV, which validate route origin authorization, and BGPsec, which was designed to provide cryptographic path validation (Gilad et al., 2017; Rodday et al., 2024). These mechanisms are not interchangeable: RPKI/ROV primarily addresses who may originate a route, while BGPsec addresses how an AS path is propagated. This distinction is central to the feature design of this Praxis.

Resource Public Key Infrastructure (RPKI)

RPKI is a specialized public-key infrastructure for Internet number resources. It allows holders of IP address space and AS numbers to create cryptographically verifiable objects that can be validated through a hierarchy anchored in the Regional Internet Registries. In routing security, the best-known RPKI object is the ROA, which expresses an authorization linking a prefix to an origin AS. RPKI therefore adds an external source of trust that is independent of the BGP UPDATE itself.

The security value of RPKI has been accompanied by extensive deployment research. Gilad et al. (2017) documented early barriers to adoption and showed that partial deployment limits global protection. Chung et al. (2019) later measured growth in RPKI deployment and examined invalid route origins, demonstrating that adoption was increasing but that operational errors remained significant. Rodday et al. (2024) synthesize the measurement literature and show that RPKI research now encompasses ROA coverage, ROV enforcement, resilience, repository behavior, and deployment incentives. Collectively, these studies establish RPKI as a mature enough measurement signal to be incorporated into routing analysis while also cautioning against treating it as complete or infallible.

RPKI also introduces infrastructure and governance considerations. Validation depends on the availability and correctness of repositories, relying-party software, certificate chains, and published objects. Heilman et al. (2014) examined transparency and trust concerns in the RPKI architecture, highlighting that security mechanisms themselves require mechanisms for accountability. These limitations are relevant to machine-learning feature engineering because an RPKI-derived feature represents the state of an external validation ecosystem at a particular time, not an absolute statement about malicious intent.

Route Origin Authorization (ROA)

A ROA is a signed authorization that specifies which AS is permitted to originate a particular IP prefix and may also constrain the maximum prefix length. The ROA does not alter BGP's route-selection process by itself; rather, it supplies validated authorization data that a relying party can use to determine whether an observed origin is consistent with the resource holder's published intent. This separation between authorization and

enforcement is important: the presence of a ROA creates evidence, whereas the routing policy of a receiving network determines how that evidence affects route acceptance.

ROA semantics are particularly useful for incident analysis because they transform an origin observation into a categorical security context. A route whose origin and prefix length are covered by an appropriate ROA can be classified as RPKI valid; a covered route that conflicts with the authorized origin or maximum length can be classified as invalid; and a route without applicable authorization is not found/unknown. Wählisch et al. (2012) showed early how RPKI information could support hijack detection, while later deployment studies demonstrate the increasing practical availability of these states.

However, invalid does not mean malicious. Xu et al. (2019) analyzed false alarms among invalid prefixes and showed that operational practices such as traffic engineering, address transfers, and aggregation behavior can produce invalid states. Chung et al. (2019) similarly found that invalid origins must be interpreted carefully. This is a strong argument for using ROA-derived states as model features rather than deterministic labels. In the present praxis, RPKI features are intended to increase discriminative context while allowing the classifier to combine that evidence with behavioral and AS-relationship features.

Route Origin Validation (ROV)

ROV is the operational process by which BGP routes are evaluated against validated RPKI origin information and then handled according to local policy. The conceptual outcome is commonly expressed as valid, invalid, or not found/unknown. ROV can substantially reduce exposure to unauthorized origin announcements when networks reject or de-preference invalid routes, but its effectiveness depends on both ROA coverage and enforcement by networks along relevant paths.

Empirical work demonstrates that measuring ROV deployment is more difficult than measuring ROA publication. Li et al. (2023) developed RoVista to measure and analyze ROV behavior, while Qin et al. (2024) investigated real-world ROV deployment and the gap between routing-security recommendations and actual practice. These studies show that enforcement is heterogeneous and can be influenced by routing policy, measurement

location, and experimental design. Thus, a route's RPKI state is directly observable from validated data, whereas whether every AS on the Internet will enforce that state is not.

For anomaly analysis, ROV- related information is valuable even when deployment is incomplete. A conflicting origin that is RPKI invalid carries different security meaning from an origin change for an uncovered prefix. Conversely, a route that remains valid does not prove that its AS path is policy compliant or that a forged-origin attack is absent. ROV therefore complements rather than replaces AS-path and relationship analysis. This complementary relationship is one of the principal reasons the present study combines RPKI-derived features with CAIDA AS-relationship features.

BGPsec

BGPsec extends BGP with cryptographic protection for the AS path. RFC 8205 specifies a BGPsec_PATH attribute containing signatures that allow a receiving BGPsec speaker to verify that each participating AS authorized propagation to the next AS in the path (Lepinski & Sriram, 2017). In conceptual terms, BGPsec addresses a limitation left by origin validation: a route can have a legitimate origin yet still traverse a manipulated or unauthorized path.

Research on secure path validation has repeatedly confronted incremental-deployment and performance challenges. Lychev et al. (2013) examined BGP security under partial deployment and found that security benefits depend strongly on routing policies and adoption patterns. Other work has explored ways to reduce cryptographic or memory overhead, reflecting the practical cost of path validation. The continuing gap between standardized BGPsec and broad operational deployment helps explain why passive analysis of AS paths and relationship semantics remains important in contemporary detection systems.

BGPsec is therefore relevant to this praxis primarily as a conceptual boundary. The study does not attempt cryptographic path validation. Instead, it uses observed AS paths, inferred AS relationships, and RPKI origin states as features that approximate different dimensions of routing legitimacy. This approach cannot provide BGPsec's cryptographic

guarantees, but it is compatible with historical RouteViews data and can be applied retrospectively across incidents where BGPsec information is unavailable.

### Current Deployment and Challenges

The routing-security literature consistently describes deployment as an incremental process. RPKI coverage has grown substantially, but ROA creation, ROV enforcement, and operational correctness remain uneven (Chung et al., 2019; Rodday et al., 2024; Qin et al., 2024). BGPsec deployment is considerably more limited. These realities mean that Internet routing currently operates in a mixed environment in which some prefixes have cryptographically verifiable origin authorization, some networks enforce invalid-route policy, and much path validation still depends on conventional routing policy and monitoring.

Several challenges follow from this partial deployment. First, uncovered routes provide no RPKI origin verdict beyond not found. Second, invalid routes can arise from benign operational mistakes. Third, origin validation does not detect every path manipulation or route leak. Fourth, measurement of enforcement is difficult because routing decisions are distributed and policy dependent. Finally, security data change over time, so retrospective studies must align RPKI state with the event period as carefully as practical.

These limitations do not reduce the value of RPKI features; instead, they define how those features should be interpreted. A machine-learning model can treat valid, invalid, and unknown states as evidence whose predictive importance is learned from labeled incidents. This use is consistent with the literature's finding that routing-security signals are informative but context dependent. This also allows the Praxis to evaluate whether adding RPKI validation features improves classification and incident-cause ranking beyond what can be achieved using behavioral BGP and AS-relationship features alone.

## AS Relationships, Routing Policies, and Valley-Free Routing

BGP routes are shaped not only by protocol mechanics but also by economic relationships among ASes. These relationships determine which routes an AS prefers and which routes it exports to neighbors. As a result, AS-path interpretation requires policy semantics. The literature on AS relationships provides a second external knowledge

source, alongside RPKI, that can enrich BGP telemetry with information relevant to route leaks and anomalous propagation.

## Autonomous System Relationships

Gao (2001) established a foundational framework for inferring AS relationships from observed BGP paths. The work formalized the idea that AS links can be interpreted according to business roles such as customer-provider, peer-to-peer, and sibling relationships. This abstraction is essential because an AS path is not simply a sequence of nodes; it is a sequence of commercial and routing-policy relationships that constrain plausible propagation.

Later research refined relationship inference as the Internet became more complex. Luckie et al. (2013) proposed relationship and customer-cone inference without relying exclusively on maximizing valley-free paths, and Giotsas et al. (2014) examined complex AS relationships that do not fit neatly into the traditional categories. These studies show that inferred relationship data are an approximation of a dynamic commercial system, but they also demonstrate that relationship inference is sufficiently informative for topology research, routing analysis, and validation.

## Customer-Provider and Peer Relationships

In a customer-provider relationship, the customer pays the provider for transit and generally expects the provider to carry traffic toward the broader Internet. In a peer relationship, two ASes exchange traffic for their respective customers without providing unrestricted transit to one another. These economic distinctions produce characteristic export policies. A route learned from a customer can generally be exported broadly because carrying customer traffic generates revenue; a route learned from a provider or peer is normally exported only to customers to avoid providing unpaid transit.

This policy structure gives AS paths directional meaning. A sequence of provider-to-customer and customer-to-provider transitions can indicate whether a route follows an economically plausible propagation pattern. For machine learning, counts and ratios derived from relationship types can therefore encode information that ordinary AS-path

length cannot. A path of length five containing a policy violation may be more suspicious than a path of length eight that follows expected relationships.

## Valley-Free Routing

The valley-free model captures a widely used abstraction of economically rational BGP propagation. In a simplified valley-free path, traffic may travel from customer to provider zero or more times, optionally traverse a peer link, and then travel from provider to customer zero or more times. A path that descends toward a customer and subsequently climbs toward another provider creates a 'valley' and can indicate that an AS is providing transit contrary to the expected relationship policy.

Valley-free routing is a common routing principle, but real-world BGP routes do not always follow it. Giotsas and Zhou (2012) analyzed valley-free violations using BGP community data and showed that real routing can contain exceptions and complexities. Complex business relationships, partial transit, route-server behavior, and imperfect inference can all create apparent violations. Nevertheless, the concept remains highly useful for route-leak analysis because leaks frequently manifest as propagation beyond the scope implied by provider and peer relationships.

For this praxis, valley-free violations are therefore treated as statistical features rather than hard rules. This mirrors the treatment of RPKI invalidity: both are semantically rich signals with possible benign exceptions. Their value lies in how they shift the probability of competing incident classes when combined with other observations.

## CAIDA Relationship Inference

CAIDA operationalizes decades of AS-level measurement research by publishing inferred AS relationships and related topology datasets. These data provide a practical way to map adjacent AS pairs in observed BGP paths to relationship categories. The methodology is grounded in research such as Gao (2001), Luckie et al. (2013), and Giotsas et al. (2014), while recognizing that no inference system has complete visibility into confidential interconnection agreements.

CAIDA relationship data are particularly suitable for retrospective BGP studies because they can be joined with AS_PATH observations without requiring access to private router configuration. Once path adjacencies are mapped to relationship types, event-level variables can be derived, including counts of customer-provider and peer links, unknown relationships, policy-transition patterns, and valley-free violations. This creates a bridge between global public telemetry and policy-aware feature engineering.

The use of inferred relationships also requires methodological caution. Relationship snapshots should be aligned as closely as possible to the incident period, and unknown or ambiguous relationships should be preserved rather than forced into a category. The resulting features represent inferred policy context, not authoritative contracts. This limitation is acceptable for statistical modeling provided it is explicitly acknowledged and evaluated through feature importance and ablation analysis.

Policy Violations and Route Leaks

Route leaks provide the clearest operational connection between AS relationships and anomaly detection. RFC 9234 describes leaks as propagation that violates assumptions about AS relationships, such as advertising a provider-learned route to another provider or peer. Because the route origin may be valid, RPKI origin validation alone cannot reliably identify these events. Relationship-aware mechanisms are therefore necessary to reason about propagation scope.

Autonomous System Internet Registry Inference for path Authorization (ASIRIA) uses inferred AS relationship information to detect and protect against leaked routes, demonstrating that policy semantics can provide practical detection value (Bagnulo et al., 2022). RoLL uses AS-triplet features to locate route leaks in real time, further supporting the importance of local relationship patterns in identifying the point at which expected propagation breaks (Xie et al., 2023). LeakFocus extends the problem from leak detection toward perpetrator localization using features that characterize valley paths and neighboring AS impact (Xie et al., 2025).

These studies strongly support the inclusion of AS-relationship features in the present model. They also expose an important gap: most relationship-aware systems are

specialized for route leaks. A general BGP incident-ranking model should retain those leak-sensitive features while allowing them to compete with origin-validation and behavioral features associated with hijacks and session resets. This broader multi-class use of relationship semantics is a central element of the study's positioning.

## BGP Anomaly Detection Techniques

BGP anomaly detection has progressed from statistical monitoring and handcrafted rules to machine-learning and topology-aware systems. The central objective of most work is to identify observations that differ from expected routing behavior. Although this objective overlaps with the present praxis, anomaly detection and incident-cause ranking are not identical. Detection asks whether an event is abnormal; ranking asks which incident explanation is most strongly supported after abnormal behavior is observed.

Statistical approaches exploit the fact that incidents can create changes in update frequency, withdrawal rate, path length, prefix visibility, and other aggregate measures. Such techniques are attractive because they are computationally efficient and often interpretable. Recurrence-based methods extend statistical analysis to temporal structure, and Al-Musawi (2024) applies recurrence quantification to BGP anomalies. Scott et al. (2025) frame BGP anomaly detection as a group-dynamics problem using multidimensional recurrence analysis. These approaches reinforce the value of temporal and distributional changes but generally do not supply protocol-specific incident-cause explanations.

Early supervised machine-learning work showed that BGP anomalies can be classified using engineered features. Al-Rousan and Trajković (2012) used SVM and HMM models to classify multiple anomaly types. Sanchez et al. (2019) compared machine-learning algorithms for BGP anomaly detection, while later studies investigated balanced or imbalanced datasets and alternative partitioning strategies. These studies establish that supervised learning can separate incident classes when informative features and labels are available, but they also reveal sensitivity to dataset construction and class imbalance.

Weakly supervised and unsupervised methods address the scarcity of labeled incidents. Dong et al. (2021) proposed ISP self-operated BGP anomaly detection based on weak

supervision. Putina et al. (2018) and subsequent stream-clustering work explored online detection from high-rate telemetry. Such approaches are valuable for discovering deviations without complete labels, but their outputs often require additional interpretation before they can support an operator's diagnosis.

Real-time and incident-specific systems provide another important research direction. ARTEMIS combines real-time monitoring with automated mitigation for prefix hijacking (Sermpezis et al., 2018). ASIRIA targets route leaks using relationship information (Bagnulo et al., 2022). DFOH focuses on forged-origin hijacks (Holterbach et al., 2024). RoLL and LeakFocus address route-leak localization (Xie et al., 2023; Xie et al., 2025). These systems demonstrate that domain-specific features can produce operationally useful detection, but their specialization also limits direct generalization to a multi-incident incident-cause ranking problem.

Semantics-aware systems narrow the gap between detection and diagnosis. BEAM analyzes routing anomalies with semantic context and incident-cause analysis as an explicit goal (Chen et al., 2024). BEAR focuses on event analysis and reporting, reflecting growing interest in converting raw BGP observations into higher-level incident descriptions (Li et al., 2025). This trajectory is significant: the field is moving from simple anomaly flags toward explanations, localization, and operator-facing summaries.

Topology-aware detection also challenges the assumption that manually engineered message statistics are sufficient. Hoarau et al. (2021) examined graph representations for BGP anomaly detection; Peng et al. (2021) used a graph attention framework; and Latif-Martínez et al. (2025) proposed operating on raw Internet topology. Graph methods can capture structural dependencies that are lost in aggregate features, but they generally require more complex models and may reduce direct interpretability. For an enterprise-oriented diagnostic system, this creates a tradeoff between representational richness and operational transparency.

Another recurring issue is observation bias. Public route collectors do not observe every route from every AS, and recent work questions the robustness of relying exclusively on public BGP monitors for hijack detection under adversarial conditions (Giaconia et al., 2025). Even outside an adversarial setting, collector placement and visibility affect what

a model can learn. This limitation reinforces the importance of event-level quality gates, holdout design, and conservative interpretation of model performance.

Overall, BGP anomaly-detection research supports four conclusions for this study. First, engineered BGP features contain substantial predictive information. Second, protocol-aware and topology-aware context improves upon purely statistical signals. Third, labeled datasets and class balance strongly influence reported performance. Fourth, specialized detectors do not eliminate the need for a unified multi-class diagnostic framework. The present study builds on these conclusions by combining behavioral, RPKI, and AS-relationship features within a hierarchical classification and ranking architecture.

## Machine Learning for Network Fault Detection and Diagnosis

The final methodology of this Praxis emphasizes a leakage-safe incident-level XGBoost stacking architecture, while Random Forest is retained as an important benchmark. The literature review therefore evaluates model families according to their relevance to structured, protocol-derived BGP features, incident-level grouping, ranking output, and operational interpretability.

### Statistical Learning

Statistical learning occupies the boundary between classical thresholding and machine learning. Baseline distributions, robust statistics, clustering, recurrence measures, and probabilistic models can detect deviations without requiring deep architectures. In BGP, these methods are useful because many incidents manifest as changes in count distributions, burstiness, path-length variance, or withdrawal ratios. They are also computationally attractive for large routing archives.

Romo-Chavero et al. (2025) illustrate a hybrid approach in which Median Absolute Deviation is combined with machine-learning classifiers including Random Forest and XGBoost. The work is important for this praxis because it shows that robust statistical screening and tree-based classification can be complementary rather than competing paradigms. Recurrence-based studies likewise show that temporal structure can be captured through engineered statistical descriptors without necessarily requiring end-to-end deep learning.

The limitation of statistical learning is semantic ambiguity. A statistically extreme event is not necessarily malicious, and a subtle policy violation may not generate the largest deviation. Statistical features are therefore best viewed as behavioral evidence. In the proposed model, update counts, withdrawal ratios, path instability, origin changes, and related variables serve this role, while RPKI and AS-relationship features add external semantic context.

Ensemble Learning

Ensemble tree methods are well suited to structured routing data because they model nonlinear interactions, tolerate mixed feature scales, and can capture conditional relationships without requiring the feature normalization and large sample sizes common in many neural approaches. Random Forest constructs multiple decision trees over bootstrapped samples and randomized feature subsets, reducing variance relative to a single tree. Gradient-boosting methods such as XGBoost construct trees sequentially to correct prior errors and often achieve strong predictive performance on tabular data.

BGP studies have repeatedly used or compared tree-based models. Sanchez et al. (2019) compared machine-learning algorithms for BGP anomaly detection, while more recent hybrid work evaluates Random Forest, XGBoost, Extra Trees, LightGBM, and CatBoost in combination with robust anomaly screening (Romo-Chavero et al., 2025). Graph-feature studies also commonly feed derived structural metrics into conventional classifiers, demonstrating that sophisticated feature engineering does not require a deep model to be useful (Shen et al., 2024).

Tree-based ensemble models are attractive for this Praxis because the input is incident-level tabular data composed of behavioral, AS-relationship, and RPKI/ROV features. Random Forest provides a strong baseline, while XGBoost provides a boosted-tree alternative that proved stronger in the final controlled comparison.

XGBoost remains an important baseline because boosted trees can capture complex interactions and often perform strongly on imbalanced classification tasks. Comparing Random Forest with XGBoost helps determine whether the observed results depend on a particular ensemble-learning strategy. The goal is not to establish that one algorithm is

universally superior, but to justify the selected architecture against a credible high-performance tabular baseline.

The hierarchical two-stage design further reflects the structure of the diagnostic problem. A first-stage decision can separate normal or non-target behavior from incident behavior, after which a second stage can discriminate among incident classes. Hierarchical classification can reduce the burden on a single flat classifier when class boundaries differ in difficulty. In the context of BGP, the design also supports threshold optimization at the first stage while preserving class-specific ranking at the second. The literature on BGP classification establishes the feasibility of supervised multi-class learning; the present study extends that logic into an explicitly hierarchical incident-cause ranking workflow.

Deep Learning

Deep-learning approaches have been explored because BGP telemetry is sequential and high dimensional. Recurrent neural networks and LSTM variants can learn temporal dependencies from update streams, and VAE-GAN architectures have been used for general time-series anomaly detection (Niu et al., 2020). BGP-specific studies have applied recurrent models to security and outage-related signals, while temporal and representation-learning research continues to expand.

Deep models offer an important advantage when raw sequence structure contains information that is difficult to summarize manually. They can learn nonlinear temporal patterns and interactions without requiring every relevant statistic to be pre-specified. However, this advantage is accompanied by higher data requirements, more hyperparameters, greater computational cost, and reduced transparency. These tradeoffs are significant when the available labeled corpus consists of a limited number of real-world BGP incidents rather than millions of independently labeled events.

The earlier design of this praxis considered a Temporal Fusion Transformer as part of a hybrid architecture. The final methodology does not retain that component. The literature review nevertheless includes deep learning because it is a major branch of BGP anomaly research and clarifies the design choice. For the present corpus and research objectives, a

structured feature approach provides a closer match to the available labels, supports direct integration of RPKI and AS-relationship variables, and enables clearer SHAP-based explanations. Deep learning therefore serves as an important comparator in the literature rather than as a component of the final model.

### Graph Neural Networks

Graph-based learning is a natural extension of BGP research because interdomain routing is inherently relational. ASes form nodes, observed adjacencies form edges, and routing events change the structure or attributes of paths over time. Hoarau et al. (2021) examined the suitability of graph representation for BGP anomaly detection, while Peng et al. (2021) used graph attention to capture relationships among features and temporal dependencies. Latif et al. (2022) further explored the potential of graph neural networks for BGP anomaly detection.

More recent work has moved toward direct use of Internet topology. Latif-Martínez et al. (2025) argue that conventional anomaly detectors rely heavily on manually curated BGP message features and propose learning from the raw AS topology. Route-leak research also derives graph robustness and centrality features before applying machine learning (Shen et al., 2024). These studies demonstrate that topology contains discriminative information beyond simple path length.

Graph models, however, are not the only way to exploit relational information. The present study encodes AS relationships and valley-free behavior as explicit event-level features. This choice sacrifices some representational richness but improves interpretability and is compatible with a modest labeled corpus. It also makes the policy semantics directly visible in SHAP analysis. Graph neural networks therefore motivate the importance of relational structure while the final methodology adopts a more transparent engineered representation of that structure.

### Explainable AI (SHAP)

Explainability is essential when a machine-learning system is intended to support network operations rather than merely achieve benchmark accuracy. A network engineer must be able to determine whether a prediction is driven by plausible routing evidence.

Feature attribution methods provide one way to expose that relationship by estimating how individual variables contribute to a model's output.

SHAP is grounded in Shapley values from cooperative game theory and provides local and global feature-attribution explanations. For tree ensembles, efficient algorithms make SHAP practical even when many predictions must be analyzed. In BGP anomaly research, Kyryk et al. (2025) explicitly apply SHAP-based feature importance to anomaly classification, demonstrating growing interest in explainable routing models. Feature-selection studies likewise emphasize that identifying which routing variables drive performance is important for both efficiency and trust.

For this praxis, SHAP serves two purposes. First, it tests whether the model relies on routing features that are technically credible, such as withdrawal ratios, AS-path instability, origin changes, RPKI states, and valley-free violations. Second, it supports comparison among feature groups. If RPKI or AS-relationship enrichment improves classification, SHAP can help determine whether the model actually uses those new variables and for which incident classes. This is stronger evidence than reporting aggregate accuracy alone.

A high SHAP value indicates that a feature influenced the model's prediction under the learned data distribution; it does not establish that the feature caused the real-world incident. The appropriate interpretation is therefore diagnostic support: SHAP explains the model's reasoning in terms of observed evidence, while the incident labels and external event knowledge define the supervised learning target. This distinction is important to avoid overstating the causal claims of a classification and ranking system.

## Summary of Research Gaps and Study Positioning

The reviewed literature demonstrates substantial progress in BGP measurement, routing security, policy inference, anomaly detection, and machine learning. Yet the literature also reveals a set of gaps that become visible only when these research areas are considered together.

The first gap is the continuing separation between anomaly detection and operational diagnosis. Many systems determine that routing behavior is unusual, and several

accurately detect a specific incident type, but fewer systems are designed to rank multiple competing incident-cause hypotheses for an operator.

The second gap is incomplete integration of routing-security context with machine-learning features. RPKI research has developed robust methods for measuring ROA coverage, invalid origins, and ROV deployment (Chung et al., 2019; Li et al., 2023; Qin et al., 2024; Rodday et al., 2024). BGP anomaly research, however, has historically emphasized message statistics, path features, or learned representations. The opportunity is to use RPKI validation state not merely as a security control but as an explanatory feature within a multi-class diagnostic model.

The third gap concerns AS-policy semantics. Relationship inference and route-leak research show that customer-provider and peer relationships, valley-free behavior, and AS triplets are highly informative for identifying leaks (Gao, 2001; Luckie et al., 2013; Bagnulo et al., 2022; Xie et al., 2023). Yet these features are often used in leak-specific systems rather than combined with origin-validation and general BGP behavioral features in a unified incident classifier. A model that evaluates hijacks, leaks, and session resets together can test whether policy semantics improve discrimination beyond their established leak-detection role.

The fourth gap is methodological reproducibility across real-world incidents. Public BGP datasets provide enormous volumes of updates but comparatively few clean, independently labeled incident events. Studies report sensitivity to class balance, vantage-point selection, and event construction. This creates a risk of high row-level accuracy that does not generalize across incidents. Event-level holdout design, quality gates, and ablation analysis are therefore necessary to demonstrate that a model has learned incident-relevant patterns rather than artifacts of a particular event window.

The fifth gap is interpretability aligned with network operations. Feature importance and SHAP-based studies are emerging, but many detection systems still emphasize precision, recall, or F1 without explaining how predictions correspond to routing semantics. For a classification and ranking system, explanation is not an optional visualization; it is part of the operational value proposition. A Network Engineer must be able to see whether a

predicted leak is supported by policy violations or whether a predicted hijack is associated with origin and RPKI evidence.

The sixth gap is the balance between model sophistication and deployability. Deep-learning and graph-based approaches have been applied to BGP anomaly detection and can learn complex representations from routing data (Peng et al., 2022; Yang et al., 2022). However, more complex models are not necessarily better when the available data are structured and represented through engineered features. Studies comparing machine-learning methods have shown that tree-based models remain highly competitive for tabular datasets and can outperform more complex deep-learning approaches in many cases (Fernández-Delgado et al., 2014; Grinsztajn et al., 2022). Random Forest and gradient-boosted trees are particularly well suited to structured feature spaces and also provide practical advantages for feature analysis and model interpretation (Breiman, 2001; Chen & Guestrin, 2016). This gap motivates the use of ensemble models in this Praxis, where BGP behavioral, AS-relationship, and RPKI information are represented as structured features and where both prediction performance and operational usability are important.

This study is positioned at the intersection of these gaps. It constructs a multi-source corpus of real-world BGP incidents, derives behavioral features from RouteViews UPDATE data, enriches those observations with CAIDA AS-relationship features and RPKI validation features, and evaluates a hierarchical two-stage Random Forest architecture against an XGBoost baseline. The model produces class probabilities that can be interpreted as a ranking of likely incident causes, and SHAP is used to expose the feature evidence underlying those predictions. The contribution is therefore not a new cryptographic routing protocol or a replacement for specialized hijack/leak detectors; it is an explainable diagnostic layer that integrates several established forms of routing evidence into a common incident-cause ranking framework.

## Literature Synthesis and Model Justification

The literature shows that different types of information can be useful when analyzing BGP incidents. The first type is behavioral information taken directly from BGP

UPDATE messages. Examples include announcements, withdrawals, AS-path length, origin changes, prefix activity, and other measures of routing instability. These features describe what changed in the routing system and have been widely used in BGP measurement and anomaly-detection research (Al-Rousan & Trajković, 2012; Fonseca et al., 2019; Putina et al., 2018). The second type of information comes from AS relationships and routing policies. Customer-provider and peer relationships, valley-free routing behavior, and other AS-path characteristics provide additional context for determining whether an observed route follows expected interdomain routing behavior (Gao, 2001; Giotsas & Zhou, 2012; Luckie et al., 2013; Bagnulo et al., 2022). The third type of information comes from RPKI and route-origin validation. RPKI provides information that can be used to determine whether an AS is authorized to originate a particular prefix (Gilad et al., 2017; Chung et al., 2019; Rodday et al., 2024). Together, these three areas provide different but complementary views of a BGP incident.

This literature provides the basis for the incremental feature design used in this Praxis. A behavioral-only model first establishes how much can be learned directly from BGP telemetry. CAIDA AS-relationship features are then added to determine whether routing-policy information provides additional value, particularly for identifying route leaks. Previous studies have shown that AS relationships and policy-aware path analysis are useful for identifying abnormal route propagation and route leaks (Gao, 2001; Luckie et al., 2013; Bagnulo et al., 2022; Xie et al., 2023). RPKI validation features are then added to determine whether origin-validation information provides additional evidence for distinguishing routing incidents. Prior research has shown the importance of RPKI and ROV for identifying unauthorized or invalid prefix-origin relationships (Chung et al., 2019; Li et al., 2023; Qin et al., 2024; Rodday et al., 2024). Comparing the model as these feature groups are added makes it possible to measure whether AS-relationship and RPKI information improve performance beyond behavioral BGP features alone.

The literature also provides support for using ensemble tree models with structured data. The final feature set in this Praxis consists of engineered numerical and categorical features rather than raw packet data or an unprocessed time sequence. Random Forest is well suited to this type of structured data because it can model nonlinear relationships

among multiple variables and has been widely used for classification problems (Breiman, 2001). Research comparing machine-learning methods has also shown that tree-based approaches remain strong choices for tabular datasets and do not necessarily require the large datasets associated with more complex deep-learning architectures (Fernández-Delgado et al., 2014; Grinsztajn et al., 2022). XGBoost provides another strong tree-based approach and is therefore used as a comparison model in this Praxis (Chen & Guestrin, 2016). These characteristics support the use of Random Forest as the primary model while still allowing its performance to be compared with another established ensemble method.

The two-stage hierarchical architecture reflects the structure of the classification problem. Instead of asking one model to make every decision at once, the first stage determines whether an observation belongs to the normal/non-target group or represents a target BGP incident. The second stage then determines the most likely incident type among the target classes. Hierarchical classification has been used in machine learning to divide complex classification problems into smaller and more focused decisions (Silla & Freitas, 2011). In this Praxis, the two-stage design also makes it possible to optimize the first-stage decision threshold separately and examine how that threshold affects false positives and missed incidents.

Evaluation of the model also needs to go beyond a single accuracy value. BGP incident data can be unevenly distributed across classes, making metrics such as precision, recall, and F1 important for understanding performance (He & Garcia, 2009; Sokolova & Lapalme, 2009). ROC and precision-recall analysis provide additional information about classification behavior at different thresholds, particularly when class distributions are uneven (Fawcett, 2006; Saito & Rehmsmeier, 2015). Because this Praxis produces ranked incident probabilities, calibration is also important. A probability should provide useful information about the confidence of a prediction rather than serve only as a score for selecting the largest class value (Niculescu-Mizil & Caruana, 2005; Van Calster et al., 2019). For this reason, the methodology evaluates classification performance, ranking performance, threshold behavior, and probability calibration.

Model explanation is another important part of the design. Previous BGP anomaly-detection studies show that machine-learning models can identify complex routing patterns, but an operational system should also provide information about why a prediction was made. SHAP provides a method for measuring how individual features contribute to a model's prediction and can be applied to tree-based models such as Random Forest and XGBoost (Lundberg & Lee, 2017; Lundberg et al., 2020). This is useful for this Praxis because predictions can be examined in relation to routing features such as RPKI validity, valley-free violations, origin changes, withdrawal behavior, and AS-path instability. SHAP therefore provides a connection between model output and routing information that a network engineer can examine.

An important limitation identified in the literature is that BGP telemetry alone cannot prove the physical cause of a network event. Public BGP collectors observe routing behavior from a limited set of locations, AS-relationship datasets are inferred from observed routing information, and RPKI coverage is not complete (Luckie et al., 2013; Chung et al., 2019; Reuter et al., 2018; Rodday et al., 2024). In addition, many BGP anomaly studies reconstruct incidents using publicly available routing data and documented event information rather than direct access to the affected routers (Orsini et al., 2016; Fonseca et al., 2020). For this reason, incident-cause ranking in this Praxis does not mean that the model proves the physical cause of an incident. Instead, it means that the model ranks the most likely incident classes based on the BGP evidence available to it. The ranked output is intended to help an engineer decide what to investigate first and does not replace router logs, configuration information, SNMP data, or other operational evidence.

The final design of this Praxis therefore brings together several established areas of research. Foundational BGP studies define the routing information that can be observed. RouteViews and related measurement platforms provide historical BGP data for analysis. AS-relationship research provides routing-policy information. RPKI provides route-origin authorization information. BGP anomaly-detection research shows that engineered routing features can be used for classification, while ensemble learning provides models that are well suited to the resulting structured feature set. SHAP adds a way to explain

how those features affect individual predictions. The methodology described in Chapter 3 brings these components together through incident corpus construction, BGP feature extraction, CAIDA AS-relationship enrichment, RPKI validation enrichment, hierarchical classification, incremental model and feature comparison, threshold optimization, calibration analysis, and inference-latency testing.

In summary, the literature provides the foundation for the main idea of this Praxis: BGP incident diagnosis can benefit from combining several forms of routing evidence rather than relying on one type of indicator. Behavioral BGP features describe what changed, AS-relationship features provide information about whether the observed path is consistent with expected routing policy, and RPKI features provide information about whether the prefix origin is authorized. This Praxis combines these sources of information in a hierarchical and explainable ensemble model and evaluates whether they improve the ranking of likely BGP incident causes for enterprise network operations.

# Chapter 3-Methodology

## Research Design Overview

This Praxis used a quantitative experimental design. Historical BGP incidents were represented as five-minute RouteViews observation windows, enriched with routing-policy and route-origin validation context, and evaluated under incident-level partitions. The independent experimental unit was the incident, not the individual BGP UPDATE message or five-minute feature row.

Figure 3.1 Final Research Workflow

Labeled incident roster -> incident-level partitioning -> RouteViews MRT acquisition -> five-minute window construction -> Behavioral features -> CAIDA enrichment -> RPKI/ROV enrichment -> V4 feature corpus -> grouped TRAIN development -> Validation confirmation -> TRAIN-only K1 development -> sealed Final Holdout evaluation.

## Incident Corpus and Partitions

The authoritative frozen v2 incident corpus contains 1,419 independent incidents: 473 Hijack, 473 Leak, and 473 Outage incidents. Confirmed duplicate groups were removed before freezing the roster. The final partitions were generated at the incident level and are authoritative for all downstream windows and feature rows.

Table 3.1 Incident Partition Summary

| Partition | Hijack | Leak | Outage | Total | Role |
|---|---|---|---|---|---|
| TRAIN | 331 | 331 | 331 | 993 | Development, grouped CV, model selection |
| Validation | 71 | 71 | 71 | 213 | Single predeclared confirmation for Phase J candidate |

| | | | | | |
|---|---|---|---|---|---|
| Final Holdout | 71 | 71 | 71 | 213 | One-time terminal evaluation |

The split design follows two methodological rules that recur throughout the pipeline. First, complete incidents are the experimental unit. Second, every downstream five-minute feature row inherits the partition of its parent incident. This prevents a long incident from contributing windows to both training and evaluation and protects the Final Holdout from indirect tuning.

The v2 roster remediation process addressed duplicate groups, timestamp semantics, leak-prefix availability, and provenance before freezing the final corpus. Confirmed duplicates were collapsed logically, replacement incidents were selected from larger candidate pools, and the final roster was audited before the v2 split was produced.

## BGP UPDATE Data Collection and Windowing

RouteViews MRT UPDATE archives were used as the public BGP observation source. Incident acquisition windows were derived from incident-specific time bounds and active-window policies. For hijack and leak incidents, detector first-to-last spans could contain silent gaps, so active windows were derived from detector-supported evidence rather than assuming the full span was continuously active. Outage intervals used documented Cloudflare start and end times when they passed quality controls.

Raw UPDATE data were parsed into observations, then aggregated into five-minute windows. Five-minute windows were selected because RouteViews archives and prior BGP measurement practice support this cadence, while still providing enough temporal structure for incident-level summarization.

A compact enrichment-evidence store was introduced so that repeated prefix, origin, and AS-path observations could be aggregated without retaining the full raw parsed corpus. Each evidence row represents a unique window, prefix, origin ASN, normalized AS path, and collector combination with an observation count preserving multiplicity. This design

allowed CAIDA and RPKI/ROV enrichment to be recalculated without reparsing all raw MRT files.

## Feature Engineering

The final V4 feature representation contains 74 features. Behavioral features quantify UPDATE volume, withdrawals, announcements, AS-path properties, prefix visibility, origin changes, and baseline deviations. CAIDA features add AS-relationship and valley-free context. RPKI/ROV features add historical route-origin authorization and validation status.

Table 3.2 Frozen Feature Families

| Feature family | Count | Purpose |
|---|---|---|
| Behavioral BGP | 44 | Temporal routing behavior from five-minute RouteViews windows |
| CAIDA AS relationship | 14 | Provider/peer/customer and valley-free path context |
| RPKI/ROV | 16 | Route-origin authorization and validation context |
| Total V4 | 74 | Final frozen model-ready feature representation |

Table 3.3 Feature Variant Definitions

| Variant | Feature count | Definition |
|---|---|---|
| V1 | 44 | Behavioral features only |
| V2 | 58 | Behavioral plus CAIDA features |
| V3 | 60 | Behavioral plus selected RPKI features |
| V4 | 74 | Behavioral plus CAIDA plus full RPKI/ROV features |

Behavioral features were frozen first so that later enrichment did not change the original routing-behavior representation. CAIDA enrichment added relationship-aware path

context, while RPKI/ROV enrichment mapped each prefix-origin observation to the historical authorization state appropriate for the observation time. The V4 schema was then frozen at exactly 74 features.

Feature extraction was intentionally separated from modeling. No Validation or Final Holdout information was used to choose the V4 feature schema, imbalance treatment, K1 architecture, or final model hyperparameters.

## Normal Label Methodology

Normal-window construction was treated conservatively. Only windows that satisfied rule-based stability criteria and did not overlap verified incident windows were considered candidate Normal examples. The Normal-vs-Incident Stage 1 experiments remained unresolved and auxiliary; the final K1 architecture does not depend on a production Stage 1 threshold.

## Model Development Protocol

TRAIN data supported iterative model development through grouped cross-validation. Grouping by incident prevented windows from the same incident from crossing folds. Validation was opened once in Phase J for a predeclared refined XGBoost candidate. K1 was subsequently developed as a versioned TRAIN-only post-Validation experiment and was not evaluated on Validation.

Table 3.4 Modeling Phases

| Phase | Purpose | Partition access |
|---|---|---|
| Baseline and imbalance phases | Establish RF/XGB baselines and I0/I1/I2 weighting effects | TRAIN only |
| Feature ablation | Evaluate V1-V4 and feature-family contribution | TRAIN only |
| Phase J | Single predeclared Validation confirmation of refined XGBoost | TRAIN + Validation |
| K1/K2/K2.5/L0 | TRAIN-only incident-level stacking, stability checks, architecture comparison, final | TRAIN only |

| | | |
|---|---|---|
| | freeze | |
| Phase L | One-time terminal Final Holdout evaluation | Final Holdout only after freeze |

## Final K1 Architecture

The terminal model is the A0_XGB_TO_XGB K1 architecture. Level 1 uses XGBoost window-level probability models. Those probability outputs are summarized at the incident level using temporal and probability-distribution meta-features. A second-level XGBoost model then classifies and ranks the incident category. The frozen second-level model hash is 40df040321ef.

Figure 3.2 Final K1 Incident-Level Stacking Architecture

Five-minute V4 feature windows -> Level-1 XGBoost probability outputs -> incident-level temporal/probability meta-features -> Level-2 XGBoost classifier -> ranked Hijack/Leak/Outage output.

## Evaluation Metrics

The primary incident-level metrics were Macro F1, Top-1 accuracy, Top-2 accuracy, and mean reciprocal rank. Per-class precision, recall, and F1 were reported for Hijack, Leak, and Outage. Final Holdout uncertainty was estimated at the incident level; window-level uncertainty was not mixed with incident-level uncertainty.

## Evaluation Integrity and Holdout Protection

The Final Holdout was sealed during feature selection, model selection, threshold experiments, architecture comparison, balancing decisions, and hyperparameter tuning. No post-Holdout model changes were made. This protects the terminal result as the study's independent generalization estimate.

Table 3.5 Evaluation-Integrity Controls

| Risk | Control |
|---|---|
| Incident/window leakage | Incident-level partitions and grouped cross-validation |

| | |
|---|---|
| Validation overuse | Single predeclared Phase J confirmation |
| Post-Validation model development | Versioned TRAIN-only K1/K2/K2.5/L0 record |
| Holdout tuning | Final Holdout remained sealed until Phase L |
| Window-count dominance | Incident-aware weighting and incident-level metrics |

## Methodological Limitations

The methodology relies on public RouteViews collectors, so visibility may differ from enterprise-local routing state. Incident labels originate from external sources and inherit their documentation limits. Leak events do not always include exact affected prefix lists in the source data, but they remain identifiable by time, leak ASN, AS-path segment, and origin/path evidence. Stage 1 Normal-vs-Incident discrimination remained auxiliary.

# Chapter 4-Results

## Overview of Experimental Results

Chapter 4 reports the evidence trail from baseline modeling through the terminal Final Holdout evaluation. The results should be interpreted in chronological order: TRAIN grouped-CV development first, one-time Validation confirmation second, post-Validation TRAIN-only K1 development third, and sealed Final Holdout evaluation last.

Table 4.1 Model Performance Progression

| Stage | Evaluation design | Feature set | Incident Macro F1 | Status |
|---|---|---|---|---|
| RF baseline | TRAIN grouped CV | Behavioral V1 | 0.463174 | Historical baseline |
| Refined RF H2 | TRAIN grouped CV | V4/I1 | 0.784588 | RF benchmark |
| Refined XGBoost I2 | TRAIN grouped CV | V4/I1 | 0.805678 | Promoted to Validation |
| Phase J Validation | Validation confirmation | V4/I1 | 0.713904 | Passed floors |
| K1 stacker | TRAIN nested CV | V4/I1 | 0.934666 | Post-Validation TRAIN development |
| K2 multi-seed K1 | TRAIN nested CV | V4/I1 | 0.934304 ± 0.002309 | Forensic/stability |
| L0 fixed deployable K1 | TRAIN nested CV | V4/I1 | 0.931653 ± 0.002150 | Pre-Holdout frozen |
| Final Holdout | One-time terminal | V4/I1/K1 | 0.920177 | Terminal |

## Feature Variant and Feature-Family Evidence

Feature-variant and feature-family experiments supported the value of adding routing-policy and origin-validation context to behavioral BGP features. RPKI/ROV features contributed strongly to Hijack classification, while CAIDA and peer-visibility context

provided useful signal for Leak and Outage separation. These are predictive feature-contribution findings, not causal claims about the underlying incidents.

## Random Forest and XGBoost Model Selection

Random Forest remained an important benchmark, but refined XGBoost outperformed the refined Random Forest comparator under TRAIN grouped-CV. The refined Random Forest reference produced incident Macro F1 of 0.784588, while the refined XGBoost candidate produced incident Macro F1 of 0.805678.

Table 4.2 Refined RF and Refined XGBoost TRAIN Grouped-CV Comparison

| Model | Incident Macro F1 | Top-1 | Top-2 | MRR | Disposition |
|---|---|---|---|---|---|
| Refined Random Forest | 0.784588 | 0.787513 | 0.932931 | 0.882578 | Benchmark |
| Refined XGBoost | 0.805678 | 0.807049 | 0.934743 | 0.892649 | Promoted to Phase J |

## Phase J Validation Confirmation

Phase J was the first authorized access to the Validation partition. The predeclared refined XGBoost candidate passed the frozen Validation floors, but the later K1 architecture was not evaluated on Validation. Therefore, Validation evidence confirms the pre-K1 refined XGBoost candidate and does not make K1 a Validation-confirmed model.

Table 4.3 Phase J Validation Metrics

| Metric | Validation value |
|---|---|
| Incident Macro F1 | 0.713904 |
| Top-1 | 0.718310 |
| Top-2 | 0.901408 |
| MRR | 0.842723 |
| Hijack F1 | 0.634146 |
| Leak F1 | 0.774648 |

| | |
|---|---|
| Outage F1 | 0.732919 |

## K1 Incident-Level Stacking Results

The K1 architecture materially improved incident-level classification and ranking compared with mean probability aggregation by allowing the second-level model to learn temporal and probability-distribution summaries across windows. K2 multi-seed stability and K2.5 architecture comparisons supported keeping the XGBoost-to-XGBoost K1 architecture.

Table 4.4 K1 Stability and Final Pre-Holdout Reference

| Experiment | Incident Macro F1 | Std | Interpretation |
|---|---|---|---|
| K2 multi-seed K1 stability | 0.934304 | 0.002309 | TRAIN-only stability evidence |
| Fixed deployable K1 | 0.931653 | 0.002150 | Frozen pre-Holdout reference |

## Final Holdout Performance

The Final Holdout was accessed once after the K1 deployable model was frozen. It contained 213 incidents, balanced across 71 Hijack, 71 Leak, and 71 Outage incidents, and 14,906 V4 window rows. The final result is the primary independent generalization estimate for this Praxis.

Table 4.5 Final Holdout Performance

| Metric | Value | Sample unit |
|---|---|---|
| Incident Macro F1 | 0.920177 | Incident |
| 95% CI for Macro F1 | [0.880117, 0.954402] | Incident bootstrap |
| Top-1 Accuracy | 0.920188 | Incident |
| Top-2 Accuracy | 0.990610 | Incident |
| Mean Reciprocal Rank | 0.958529 | Incident |

Table 4.6 Final Holdout Per-Class F1

| Class | F1 |
|---|---|
| Hijack | 0.877698 |
| Leak | 0.904110 |
| Outage | 0.978723 |

Table 4.7 Final Holdout Ranking Distribution

| True-class rank | Incident count |
|---|---|
| Rank 1 | 196 |
| Rank 2 | 15 |
| Rank 3 | 2 |
| Top-2 total | 211 of 213 |

## Class-Specific Interpretation

Outage was the strongest final class, with F1 of 0.978723. Leak achieved F1 of 0.904110, indicating strong separation after Behavioral, CAIDA, and RPKI/ROV enrichment. Hijack remained the most difficult final class, with F1 of 0.877698, consistent with the complexity of distinguishing unauthorized origin behavior, sparse visibility, and overlapping path changes.

## Runtime and Operational Feasibility

The measured warm K1 downstream classification path was approximately 25.58 milliseconds per incident after features and Level-1 probabilities were available. This measurement should not be described as end-to-end live detection time because it excludes live BGP stream ingestion, parsing, feature construction, and production integration. It supports the narrower claim that the trained downstream ranking step is computationally lightweight.

## Negative and Null Findings

Several negative results strengthened the final design. Stage 1 Normal-vs-Incident discrimination did not become the final architecture because Normal precision/recall tradeoffs remained weak. Domain-expert model variants were not adopted. Class balancing alone was not sufficient. Threshold tuning was not sufficient. These results reduced the candidate space and made the final K1 selection more defensible.

## Chapter Summary

The terminal evidence shows strong incident-level generalization on the untouched Final Holdout, with high ranking quality and especially strong outage performance. The strongest claim supported by the evidence is predictive classification and ranked incident-cause output from public BGP routing telemetry, not causal proof or enterprise-local root-cause diagnosis.

# Chapter 5-Conclusions

## Overview

This Praxis developed and evaluated a reproducible machine-learning pipeline for classifying and ranking BGP hijack, leak, and outage incidents. The final system combined behavioral BGP features, CAIDA AS-relationship context, and RPKI/ROV route-origin validation features in an incident-level XGBoost stacking architecture.

## Mapping of Findings to Research Questions and Hypotheses

Table 5.1 Research Question Findings

| Research question | Finding | Disposition |
|---|---|---|
| RQ1 | Historical RouteViews data supported a reproducible incident corpus and V4 feature representation. | Supported |
| RQ2 | CAIDA and RPKI/ROV enrichments provided useful predictive signal, with RPKI especially relevant to Hijack. | Supported with caveat |
| RQ3 | K1 XGBoost-to-XGBoost stacking was selected after TRAIN-only architecture comparison and forensic checks. | Supported |
| RQ4 | The locked K1 model generalized strongly on the untouched Final Holdout. | Supported |

## Key Contributions

- A reproducible BGP incident feature-engineering pipeline using historical RouteViews MRT data.
- A frozen 74-feature Behavioral+CAIDA+RPKI/ROV representation for hijack, leak, and outage classification.

- Incident-aware partitioning, grouping, and weighting controls that address unequal window counts and leakage risk.
- A leakage-safe incident-level temporal stacking architecture that produces both class predictions and ranked incident-cause output.
- A terminal one-time Final Holdout evaluation on 213 independent incidents.

## Limitations

- Public RouteViews collectors provide partial interdomain visibility and may not reflect enterprise-local routing state.
- Incident labels come from external incident sources and inherit their documentation quality limits.
- Normal-vs-Incident Stage 1 discrimination remained unresolved and auxiliary.
- K1 was developed after Validation, so the Final Holdout is the primary independent generalization estimate for the terminal K1 model.
- The Final Holdout contains 213 incidents; confidence intervals should be interpreted with that sample size in mind.
- The measured 25.58 ms downstream K1 runtime excludes live ingestion, parsing, and feature computation.

## Recommendations for Future Research

- Develop a stronger real-time Normal-vs-Incident formulation using streaming incremental baselines and operator-validated benign periods.
- Evaluate shorter observation horizons such as 30 seconds, 1 minute, 2 minutes, and 3 minutes after rebuilding compatible features from higher-resolution evidence.
- Integrate enterprise-local telemetry, including router logs, BMP feeds, SNMP, flow telemetry, and operator ticket outcomes, to connect public BGP symptoms with local operational context.
- Extend the corpus with additional verified incidents, collectors, and geographic perspectives.

- Study calibrated ranked probabilities and operator diagnostic-time impact through controlled user studies.
- Explore graph-based AS representations and sequence models once larger leakage-safe incident corpora are available.

## Conclusion

The final evidence supports the central claim that BGP incident classification and ranked incident-cause output can be learned from a carefully controlled combination of RouteViews behavior, AS-relationship context, and RPKI/ROV signals. The terminal K1 system achieved Macro F1 of 0.920177 and ranked the true class in the Top-2 for 211 of 213 Final Holdout incidents. These findings support the Praxis as an engineering contribution to BGP incident analysis while preserving clear boundaries around causality, public-collector visibility, and unresolved Normal detection.

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